Who Believes and Who Shares Fake News: A Multi-Agent System Application for Experimental Misinformation Research with LLMs

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August 28, 2025

Abstract

In the digital age, the proliferation of misinformation poses significant challenges to democratic discourse, particularly in electoral contexts. This paper creates a one-to-many multi-agent system (MAS) prototype with LLM agents based on anonymized survey respondent data from Taiwan. The simulated pseudo-lab environment allows us to evaluate the efficacy of real-world misinformation without exposing human subjects to harmful content. From our artificial experiment, we find that agent respondents with strong pro-China attitudes demonstrate greater susceptibility to misinformation, particularly towards content fostering US skepticism. Yet this effect is largely reduced after exposure to a fact-checking intervention, suggesting that information literacy strategies can be effective even in polarized contexts. Nevertheless, we maintain a conservative interpretation of the effectiveness of fact-checking interventions in our artificial experiment.

Keywords: Large language models, generative AI, multi-agent simulation, misinformation, fake news, Taiwan election, experimental design.

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1 Introduction

Advances in generative artificial intelligence (AI) technology like large language models (LLMs) have afforded researchers with the capacity to simulate authentic human responses across various contexts (i.e., Aher et al., 2023; Park et al., 2023; Ma et al., 2024), including replicate human subject studies in economics, psycholinguistics, and social psychology. For example, Park et al. (2024) found that generative agents constructed based on real participant data can replicate the same participants' responses in a survey with 85% accuracy.

With generative agents exhibiting remarkably human-like traits - from nuanced reasoning and self-reflection to strategic planning (Yao et al., 2022; Jiang et al., 2024; Bisbee et al., 2024; Yu et al., 2024; Aher et al., 2023) — there is potential to tap on generative agents to simulate human behavior in ethically challenging research domains such as misinformation. [1] Previous experimental studies have made significant contributions to understanding misinformation dynamics, particularly in identifying individuals' susceptibility to false information (Maertens et al., 2021; Rathje et al., 2023; Pennycook et al., 2020) and determining the effectiveness of fact-checking and debunking interventions (Chan et al., 2017; Roozenbeek et al., 2024; Lewandowsky and van der Linden, 2021; Thorson, 2015).[2] However, serious questions remain over the behavioral consequences of misinformation exposure (Murphy et al., 2023).

In this paper, we demonstrate that generative AI can be used in a tailored behavioral experiment to help us evaluate the efficacy of misinformation content without subjecting human participants to potential harm. As a concrete application of the proposed methodology, we focus on the case of Taiwan in the lead-up to the 2024 presidential elections, a democratic setting where misinformation is relatively widespread, especially during elections. Building upon previous multi-agent frameworks (i.e, Gao et al., 2023; Ross et al., 2019; Du et al., 2024; Aher et al., 2023), we implement an experimental framework that uses LLMs as agents to simulate human respondents based on anonymized survey respondent data from Taiwan. Specifically, our experimental design tests how AI agent

[1]What distinguishes misinformation from disinformation is that while the former refers to the unintentional sharing of false information, the latter involves deliberate deception (Lim and Donovan, 2021). As intentionality is hard to prove, and coordinated influence campaigns can be hard to trace, for the purposes of this paper, we use the term 'misinformation'.

[2]We note some distinguish between fact-checking and debunking, with the former perceived as more impartial and non-partisan than the latter (Pamment, James and Kimber-Lindwall, Anneli, 2021). For the for the purposes of this paper, we use both terms interchangeably.

respondents with various predispositions towards mainland China respond to different misinformation content and fact-checking interventions.

Unlike previous approaches in political science, we simulate an experimental environment with AutoGen to build a one-to-many multi-agent system prototype with LLMs.[3] In this system, one main agent serves as the survey manager while 1,140 agents function as AI respondents. Within our prototype, each agent operates according to its assigned persona with an LLM backend, performing independent reasoning and inferences. Importantly, we maintain strict separation between AI agent respondents, ensuring that each agent' s processing cache remains private and is not shared with other agents. This separation preserves the independence of their responses according to their ideological preferences and political behaviors derived from their corresponding survey responses.

Our findings suggest that even after controlling for region fixed effects and multiple demographic variables, agent respondents with positive predispositions towards mainland China remain significantly associated with higher misinformation susceptibility. However, this effect is substantially reduced following a debunking message intervention. We believe our misinformation prototype effectively helps us test the responses to misinformation content and fact-checking interventions at scale, without exposing human subjects to potential harm. Nevertheless, we exercise caution when drawing parallels between AI behavior and human behavior.

2 Case Studies: Two Polarized Scenarios of Misinformation During Taiwan's 2024 Election

As the country most targeted globally by foreign disinformation operations (V-Dem Institute, 2019), Taiwan is the proverbial "canary in the coal mine" that can signal emerging patterns of misinformation and foreign interference in democracies (Chang et al., 2024). Misinformation has had tangible effects on Taiwan. For example, following local elections in 2018 that dealt crushing losses to the incumbent Democratic Progressive Party, a survey found that among voters who were exposed to news articles containing false in-

[3] `AutoGen` is an open-source Python framework that facilitates collaborative interactions among multiple LLM-based agents (Wu et al., 2023; Microsoft, 2025). Its key features make it particularly suitable for our research, including the ability to create autonomous agents with distinct roles, support for building complex conversation flows, memory management for long-running interactions, and customizable agent behaviors.

formation, more than 50% had cast their votes under the impression that the articles were accurate and authoritative.

There are three main factors underpinning Taiwan's vulnerable position. The first is Chinese media manipulation, which has been characterised by Taiwanese leaders as "cognitive warfare" (Davidson, 2022). Although precise attribution is often difficult, many exposed cases of disinformation in Taiwan can be linked to the Chinese state (Rauchfleisch et al., 2023; Wang et al., 2020; Hung and Hung, 2022). The second is Taiwan's deregulated commercial media landscape, which is "dominated by sensationalism and the pursuit of profit" (Reporters Without Borders, 2024). This open information environment facilitates the spreading of propaganda and misinformation, be it through domestic pro-unification media that take instructions directly from China, profit-oriented content farms, domestic social media influencers, or Chinese state actors (Wang, 2020; Hung and Hung, 2022; Wang et al., 2020). The third is political polarization within Taiwanese society, and this is important because foreign disinformation has been found to be key in driving polarization in societies with existing high degrees of polarization (Vasist et al., 2024).

Taiwan is a polarized society where the main cleavage is not the left-right divide that we see in Western democracies, but one that relates to Taiwan's relationship with China (Huang and Kuo, 2022; Hsiao and Cheng, 2014; Sheng, 2002). This cleavage is also reflected in a polarised information environment, with media outlets arrayed along China-friendly, 'pan-Blue' lines or pro-sovereignty, 'pan-Green' lines (Hsu, 2014; Rawnsley et al., 2016). As contemporary misinformative narratives are mostly aimed at attacking the policy stances of the pro-sovereignty incumbent party, following Van Bavel et al. (2024)'s Identity-based Model of Political Belief, we posit that these narratives should appeal more to those positively predisposed towards mainland China, including supporters of the opposition Kuomingtang (KMT). Hence, the four main hypotheses of this study are as follows:

H1: Agent respondents with pro-China attitudes are more likely to rate misinformation on Taiwanese external politics as reliable.

H2: Agent respondents with pro-China attitudes are more likely to rate misinformation on Taiwanese internal politics as reliable.

H3: Agent respondents with pro-China attitudes are more likely to share misinformation on Taiwanese external politics.

H4: Agent respondents with pro-China attitudes are more likely to share misinformation on Taiwanese internal politics.

Despite Taiwan's vulnerability, the government's approach to tackling misinformation – which largely avoids heavy-handed regulation and taps on civil society to debunk and prebunk misinformation – is seen as exemplary (European Parliament, 2023). However, there is some doubt over the efficacy of fact-checking and debunking interventions in polarized democracies, because partisan behavior—and not ignorance—is found to be driving the sharing of fake news (Osmundsen et al., 2021; Reinero et al., 2024). Accordingly, we have two additional hypotheses:

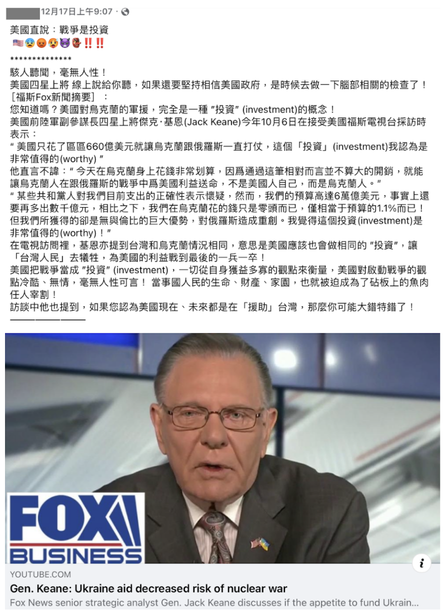
H5: Agent respondents in the treatment group (exposed to debunking message) will be more likely to revise down the credibility rating for the misinformation on Taiwanese external politics, compared to those in the control group.

H6: Agent respondents in the treatment group (exposed to debunking message) will be more likely to revise down the credibility rating for the misinformation on Taiwanese internal politics, compared to those in the control group.

To test these hypotheses, we expose agent respondents to two actual examples of misinformation that circulated during the 2024 Taiwan presidential election. These pieces of false information – one casting doubt on the intentions of the US military and the other alleging government corruption in vaccine procurement – correspond to longstanding misinformation frames that aim to undermine trust in the incumbent government's policies (Taiwan Fact Check Center, 2023b).

Table 1 shows these highly polarizing false claims as they appeared on Facebook and LINE during the 2024 election, which were fact-checked by independent organizations Taiwan FactCheck Center (2023) and MyGoPen (2023), respectively. There will be more details in the 'Experimental Design and Procedure' section as to how we sort agent respondents based on their pro-China stance, but we assess agents' responses to misinformation by asking them to evaluate the credibility of each statement and their likelihood to share it on a 5-point scale.

Table 1: Two Pieces of Misinformation on Taiwan Social Media during the 2024 Presidential Election

<table>
<tr><th>Scenario 1: US Military Misinformation</th><th>Scenario 2: Vaccine Misinformation</th></tr>
<tr><td>


</td><td>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<tr><td>

False Claims:

- Undermine Taiwan-US relations by portraying Taiwan as merely a strategic tool rather than a genuine ally
- Promote the narrative that US support is unreliable and self-serving by drawing parallels to Ukraine aid
- Leverage actual US statements about “investment” and “American interests” to create doubt about US commitment to Taiwan’s security

</td><td>

False Claims:

- Erode public trust in COVID-19 health policies and vaccine policies
- Create fear about vaccines
- Claims that the Ministry of Health and Welfare “opened backdoors” to import “low-cost” foreign flu vaccines
- Imply government-manufacturer collusion

</td></tr>
<tr><td>

Fact-Check Results (Taiwan FactCheck Center):

- The US general’s statements were quoted out of context
- Deliberately creates false equivalence between the situations in Ukraine and Taiwan

</td><td>

Fact-Check Results (MyGoPen):

- Vaccine procurement followed proper regulations; it was not a“backdoor” deal
- Both domestic and imported vaccines undergo strict vetting processes

</td></tr>
</table>

3 Multi-Agent Systems: Concepts and Implementation in Experimental Misinformation Research

To implement our misinformation MAS prototype, we use `AutoGen0.2`, an open-source Python framework developed by Wu et al. (2023) and Microsoft (2025) that facilitates collaborative interactions between the survey manager and agent respondents. Its key features make it particularly suitable for our experimental research in misinformation scenarios, including the ability to create autonomous agents with distinct demographic and political preferences based on survey data, customizable agent behaviors, support for complex conversation flows, and memory management for long-running interactions.

The `AutoGen` provides various agent architectures, each with unique features and functions: `UserProxyAgent` excels at task supervision and problem-solving, while `AssistantAgent` performs as a human-machine actor, executing code and providing human judgment based on pre-defined personas. In our prototype, we leverage these features by using `UserProxyAgent` as our survey manager to coordinate the experimental process and ensure smooth execution, while `AssistantAgent` instances simulate 1,140 respondents with unique characteristics from the real survey data, each with an independent cache store not shared with other agents, providing responses that reflect their corresponding assigned demographic and political characteristics.

Our multi-agent system use OpenAI's `GPT-4-32k Extended Model` for both survey manager and agent respondents. We selected this closed-source model primarily for its ability to efficiently orchestrate multiple agents and parallel model instances without the cache storage complications typically encountered between different agents.[4]

3.1 Implementation and Prototype Workflow

The multi-agent system requires agents capable of sophisticated interactions, including establishing distinct personas, comprehending questionnaires, performing reasoning processes, following reinforced instructions, and maintaining coherent conversations while responding to survey manager directives. Meanwhile, the survey manager operated by

[4] Additionally, running multiple LLMs requires independent, non-shared memory space to store their caches. Using open source LLMs with appropriate reasoning capabilities and sufficient token capacities demands substantial RAM for local operation. High-Performance Computing is an alternative approach that we plan to explore in future research with open-source LLMs.

an independent GPT-4 model `UserProxyAgent` executed the survey questionnaire, collected responses, and ensured agent respondents completed the questionnaire. Each process consumed an average of 16,275 tokens per agent (ranging from 8,080 to 21,881).[5] Given these demands, GPT-4's extensive context window was essential to capture the full scope of interactions between agent respondents and the survey manager, enabling us to effectively simulate and analyze the complex dynamics of misinformation engagement.[6]

Figure 1: Multi-Agent System Prototype for Misinformation Survey Experiment

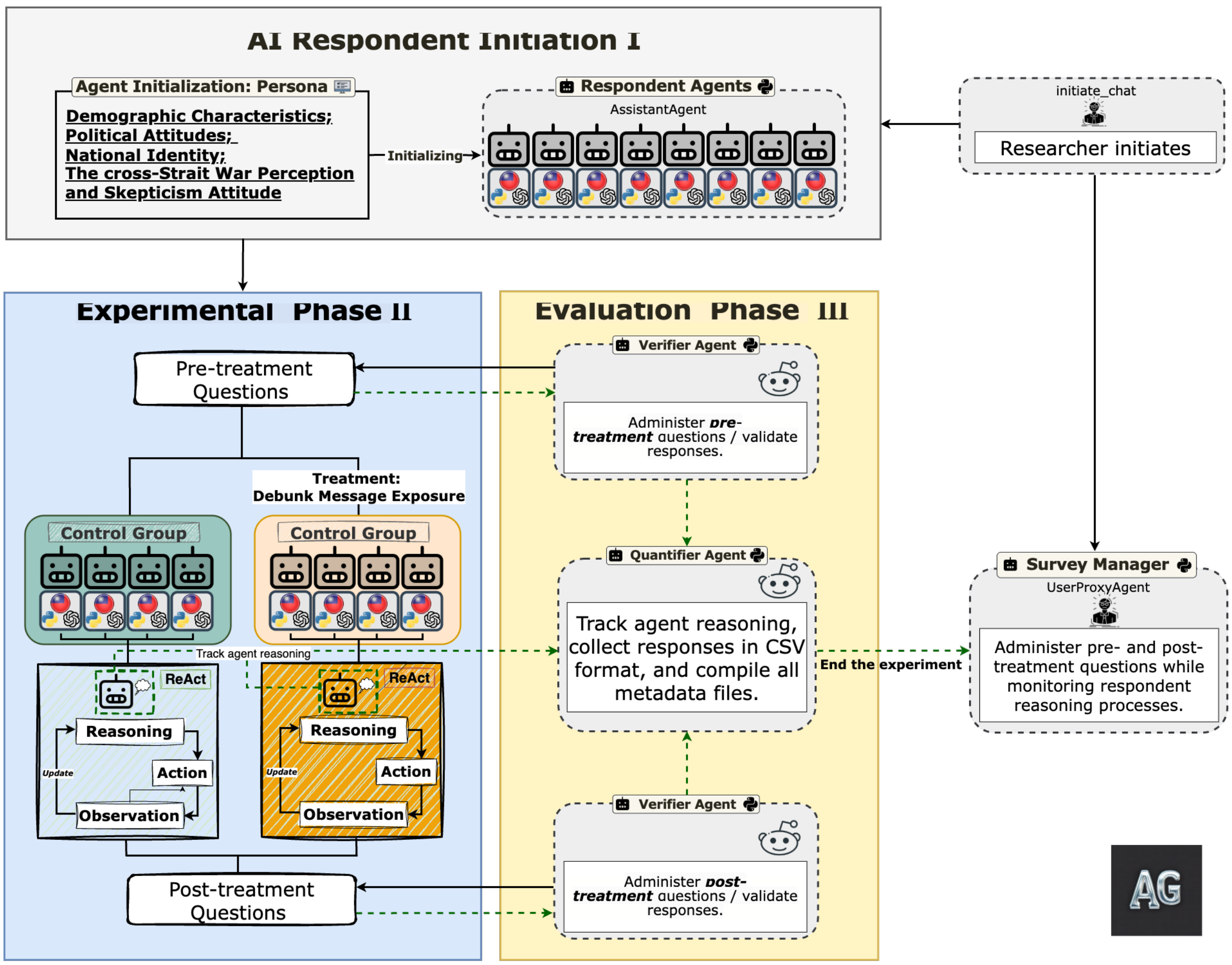


Figure 1 presents our multi-agent system for artificial survey design. The workflow consists of three main phases. In the **Construction of Agent Respondent** ■, we initialize

[5] GPT-4-32k Model has a maximum token limit of 32,768 tokens.

[6] To ensure our results are reproducible, we set the seed to 42 and temperature to 0.1, reducing randomness in model outputs and making responses more consistent in alignment with their predefined personas. Our experiment run time took a total of approximately 21 hours to complete.

1,140 agent respondents with personas derived from the survey. In the **Experimental Procedure**, AI agent respondents participate in a randomized controlled trial that includes pre-treatment questionnaires, exposure to treatment conditions with debunking information, and post-treatment surveys. The **Evaluation Architecture** employs an automated evaluation system with two key checkpoints: first validating that agent respondents provide valid responses, then ensuring they complete the entire experimental questionnaire. These three phases are managed by the Survey Manager, which orchestrates the whole experiment after being initiated by our researchers. The following sections explain how our MAS is designed and operated with LLMs.

3.1.1 Survey Manager Agent: Coordinating the Multi-Agent Experimental Framework

The Survey Manager is operated by `UserProxyAgent` in AutoGen to automatically execute the experiment. This agent is designed to simulate a survey manager during the 2024 Taiwan presidential election, capable of independently conducting our designed questionnaire without human intervention. The core design philosophy of this agent is to maintain neutrality while ensuring standardized and complete data collection. Through carefully designed system instructions, the Survey Manager agent harnesses key background information about Taiwan's political environment, including major political parties, candidates, and key election issues. In particular, when administering our experimental questions from Q1 to Q8, the agent is programmed to guide agent respondents to answer in a standardized format and monitor agent respondents' reasoning inference. In addition, we also implemented token management strategies and automatic termination mechanisms, ensuring surveys can be completed efficiently within technical limitations. To illustrate this implementation, the survey manager persona shown below demonstrates how we constructed this agent role in our multi-agent system through prompt engineering.

Prompt Engineering of Survey Manager Persona

ID: Survey Manager
Description: You are a survey interviewer conducting a survey against the backdrop of the 2024 Taiwan presidential election.
System Message:
You are a survey interviewer conducting a survey against the backdrop of the 2024 Taiwan presidential election. The main candidates for the 2024 Taiwan presidential election are DPP's Lai Ching-te and Hsiao Bi-khim, KMT's Hou Yu-ih and Jaw Shaw-kong, TPP's Ko Wen-je and Wu Hsin-ying. The election focus includes cross-strait relations, economic development, housing issues, energy policy, and other topics.
This survey is divided into two stages. the first stage (Q1-Q4) answering the questionnaire, absorbing new information in the middle, and then continuing to answer the second stage (Q5-Q8). Please ensure that respondents answer all 8 questions (Q1 to Q8).
Interview Tasks:

Please ensure respondents answer all 8 questions (Q1 to Q8):

1. Present information objectively, without suggesting correct answers
2. Accept option code responses, such as 01, 02, etc.
3. Clearly instruct respondents to use the following unified format to answer questions:
    - **Thinking:** The thought process for all questions, based on their political stance (Blue/Green/White), national identity, cross-strait and international relations perspectives, etc.
    - **Option Codes:** List option codes for all 8 questions (Q1 to Q8)
    - **Explanation:** Brief explanation of the overall selection reasons
    - **Final Answer:** Option code and brief explanation for each question
4. Maintain a neutral attitude, not favoring any political party

Token Management Strategy:

- Prioritize recording option codes for all Q1-Q8, even if explanation parts need simplification
- Avoid repeating complete option descriptions; subsequent questions can be briefly prompted
- If the conversation approaches token limit, remind respondents to answer briefly

After receiving responses, check if they match the respondent's political stance characteristics. Please ensure respondents answer according to their political stance before Taiwan's 2024 election. Even if questions 5-8 are similar to questions 1-4, please ensure respondents answer each question separately.

3.1.2 Personas and Construction of Agent Respondent

In our implementation prototype, we simulate respondent behaviors based on anonymized real-world data from the *Research on China Image Survey* conducted in Taiwan since 2014 (Wu, 2024).[7] When simulating responses to misinformation scenarios, we created authentic agent personas that genuinely represent the perspectives of Taiwanese citizens in the context of the 2024 presidential election. In our experiment, we created a total of 1,140 agent respondents. To strengthen coherent reasoning in our simulated respondents, we enhanced agent inference capabilities through the implementation of the Reasoning-Acting-Observation (ReAct) mechanism (Arabzadeh et al., 2024a,b). This implementation enabled agents to respond to misinformation exposure in a manner consistent with their pre-assigned personas and experimental conditions.[8]

Each agent respondent was created based on a total of 36 survey questions from the 2024 wave across 5 categories: 13 questions about demographics, political attitudes, and media attention; 3 questions evaluating Taiwan's 2024 presidential candidates; 3 questions on national identity; 13 questions regarding war perception, US role, and Cross-

[7]The Research on China Image Survey, directed by Dr. Chung-li Wu, is a long-term survey project conducted by the Institute of Political Science at Academia Sinica in Taiwan since 2014. The survey investigates Taiwanese public perceptions of mainland China across multiple dimensions, including ethnic identity, unification-independence issues, cross-strait relations, economic relations, trade partnerships, and educational exchanges. The 2024 wave of the survey completed a total of 3,053 respondents.(see `https://srda.sinica.edu.tw/plan/?idx=SRDA.AS027`).

[8]Our prototype implementation follows the methodology outlined by Arabzadeh et al. (2024a,b).

Strait conflicts (including China-US-Taiwan relations and US skepticism); and 4 questions on economic assessment.[9]

The example below (one from our agent respondents) illustrates how we structured each agent respondent to initialize personas in our multi-agent system. To make the prompts more natural, we systematically converted the original questionnaire into first-person format. In this structure, text with gray underlines represents the respondent's original answer choices, while parentheses and items beginning with "v-" indicate the questionnaire ID from the 2024 wave of the Research on China Image Survey. The format encapsulates comprehensive citizen attributes, including demographic information, political attitudes, identity, and perspectives on cross-strait relations and attitudes toward the United States. The original survey in Chinese and English can be found in Appendix A.1.

[9] All survey questions used to create agent personas are derived from the 2024 Research on China Image Survey, as detailed in Appendix A.1.

ID: 1010 (One of agent respondent from the survey.)
Description: New Taipei City resident, supports DPP.
System Message:
In the 2024 Taiwan presidential election, you will play the role of the following voter:

【Demographics, Political attitudes, and Media Attention (13 Items)】

I am female (v49) born in 1969 (v39). I have a technical college/university education (v43). My total family income is between NT$50,000 and NT$69,999 (v48). My household registration is in New Taipei City (v40). My father is a Taiwanese Minnan person (v42). I support the DPP (v46). My level of support for this party is moderate (v47). Overall, I am very interested in political matters (v34). I am satisfied with the overall performance of the Tsai Ing-wen government (v35). If I were to vote for president tomorrow, I would vote for Lai Ching-te (v8). If there were an election for president and legislators tomorrow, I would definitely vote (v13). I pay some attention to media coverage of US-China-Taiwan relations (v29).

【Evaluation of Taiwan's 2024 Presidential Candidates (3 Items)】

Among the four presidential candidates, using a scale where 0 means; strongly dislike, 5 means neutral, and 10 means strongly like. I rate DPP presidential nominee Lai Ching-te at 1 point (v24s1). I rate presidential candidate Ko Wen-je at 7 points (v24s2). I rate KMT presidential candidate Hou Yu-ih at 1 point (v24s3).

【National Identity (3 Items)】

I feel very proud to be Taiwanese (v36). Regarding Taiwan's future unification-independence issue, I tend to prefer maintaining the status quo first, then moving toward independence later (v38). When asked about my identity, I consider myself Taiwanese (v41).

【War Perception, US Role, and Cross-Strait Relations (13 Items)】

Regarding whether China will attack Taiwan, for the next 10-20 years, I believe China is very likely to use military force against Taiwan (v4s4); and for the next 5-10 years, I believe China is very likely to use military force against Taiwan (v4s5). I believe the United States definitely will not directly send troops to aid Taiwan (v32).

- **US-China Relations Perspective**

My overall impression of the United States is very negative (v30), and my overall impression of China is very negative (v31). If Taiwan and China were to engage in war, I believe we Taiwanese would resist (v33).

- **US-China Power Comparison**

In terms of military competition, I believe the United States is currently somewhat stronger militarily than China (v16); but in 20 years, I believe China will be somewhat stronger militarily than the United States (v17). Regarding economic influence, I believe China currently has somewhat more economic influence than the United States (v18); and in 20 years, I believe China will have much more economic influence than the United States (v19).

- **US-China Relations and Taiwan's Position**

Regarding the United States' influence on cross-strait relations, on a scale from 0 to 10, where 0 means the US causes Taiwan Strait instability and 10 means the US promotes Taiwan Strait stability, I rate it 2 points (v20).

Regarding China's influence on Taiwan, some view China as an opportunity for Taiwan's trade and economic growth, while others see it as a threat to Taiwan's national security and democratic freedoms. On a scale from 0 to 10, where 0 means 'complete threat' and 10 means 'complete opportunity,' I rate it 4 points (v21).

On the question of whether Taiwan should lean toward the United States or China, on a scale from 0 to 10 (where 0 represents leaning toward China and 10 represents leaning toward the United States), I give it 5-equidistant from both countries, therefore I believe Taiwan should remain neutral, equidistant from both countries (v22).

【Agent Itself and Taiwan Economic Assessment (4 Items)】

I believe Taiwan's current economic situation is worse compared to a year ago (v25). I feel my own current economic situation is worse compared to a year ago (v27). Regarding Taiwan's economic situation in the next year, I believe it will worsen (v26). Regarding my own economic situation in the next year, I believe it will improve (v28).

3.1.3 Agent Evaluation Architecture

To evaluate the completeness and coherence of our survey-based agents, we implemented a multi-agent evaluation framework following the AgentEval procedure (Woffinden-Luey and Kis, 2024; Arabzadeh et al., 2024b). Our system comprises two specialized evaluation agents (`VerifiedAgent` and `QuantifierAgent`) that work alongside the `Survey Manager` in our prototype. While the original `AgentEval` framework focuses on assessing the utility of LLM applications (Woffinden-Luey and Kis, 2024; Arabzadeh et al., 2024b), we adapt it specifically to monitor and validate two key aspects: 1) `VerifiedAgent` ensures all agent respondents answer and complete the experimental questionnaire, and verifies that responses follow the correct ReAct-based format. `VerifiedAgent` collects agent respondent answers and forwards them to `Quantifier-Agent`. If an agent respondent does not provide a valid response, we set up three attempts, and if unsuccessful, we skip that agent respondent and mark it as missing. 2) `QuantifierAgent` monitors the thinking processes of all agent respondents and collects and organizes the survey responses from `VerifiedAgent`, then reports to the `Survey Manager`.

3.1.4 Experimental Procedure

We randomly assigned our 1,140 agents into two balanced groups (573 control, 567 treatment) using stratified sampling based on geographic regions (Northern, Central, and Southern Taiwan), with the treatment group receiving debunking messages while the control group did not.[10]

Table 2 presents our experimental design and procedure. In the pre-treatment phase, we expose all agent respondents to false information that circulated during the 2024 Taiwan presidential election. These types of highly polarized misinformation were widely shared on Taiwanese social media platforms such as Facebook and LINE during the 2024 election period.[11] In the post-treatment questions, we exposed the treatment group to a neutral message informing them that the statements had been fact-checked and verified to be false. The manipulated variable is the presence of the debunking message: *“**This***

[10]We sampled 1,140 agent respondents based on regional proportions from the total 2024 Research on China Image Survey sample of 3,052. See Appendix C.2.

[11]LINE is a widely used instant messaging application with extremely high market share in Taiwan and East Asia, similar to WhatsApp. It has become the primary communication tool for Taiwanese citizens in their daily lives and, due to its widespread usage, serves as an important channel for rapid information spread including political content and misinformation, especially during election periods.

statement has been verified by independent fact-checkers and proven to be false news.* 该陈述已通过独立事实查核人员的核查，并被证实为假新闻。**” The control group is not exposed to any new messages. Once again, we ask the agent respondents from both groups to evaluate the credibility of each statement, and their likelihood to share these statements, on a 5-point scale. It is worth noting that to avoid potential demand effects in LLMs,[12] we deliberately retained all pre-treatment misinformation statements and questions in their original form without modification. For the treatment group, we simply added the phrase “after seeing the fact-checking results...***” at the beginning of the questions as part of the treatment condition. All questionnaires were administered in Traditional Chinese throughout our experimental prototype.[13]

[12] LLMs may exhibit a phenomenon similar to the Dunning-Kruger effect, whereby after reviewing debunking exposure, LLM-operated agent respondents might overestimate their abilities when re-answering previously examined questions. See discussion in this thread on Stack Overflow.

[13] In this paper, we use simplified Chinese characters for display due to LaTeX encoding compatibility issues, though all actual experiments were conducted using traditional Chinese characters as appropriate for the Taiwanese context.

Table 2: Experimental Design and Procedure

	Control Group	Treatment Group
No. of Agent Respondents	573	567
Pretreatment Question I	***Have you heard American former military strategist Jack Keane say that the Ukraine war is an investment? The US only spent $66 billion to make Ukraine fight with Russia... Keane then mentioned that Taiwan is the same, Taiwanese citizens are Americans' "investment" in cheap warfare. America is cold and calculating, with no real intention to help Taiwan!*** **Q1.** Please read the above statement and evaluate its credibility. (1)Very unreliable (2)Unreliable (3)Undecided (4) Reliable (5) Very reliable (96) Skip question. **Q2.** After reading this statement, how likely are you to share it on social media? (1)Very unlikely (2)Unlikely (3) Undecided (4)Likely (5)Very likely (96) Skip question.	
Pretreatment Question II	***Those who get flu vaccines annually should be careful! The Ministry of Health and Welfare is now opening backdoors again, letting Medigen import cheap foreign vaccines without even doing research? This unscrupulous government is just trying to make money, common people must keep their eyes wide open! They won't take responsibility if something goes wrong!*** **Q3.** Please read the above statement and evaluate its credibility. (1) Very unreliable (2) Unreliable (3) Undecided (4)Reliable (5) Very reliable (96) Skip question. **Q4.** After reading this statement, how likely are you to share it on social media? (1) Very unlikely (2)Unlikely (3) Undecided (4)Likely (5)Very likely (96) Skip question.	
Treatment	**[Empty]**	**Debunk Message Exposure**
Post-treatment Question I	***The same misinformation statement shown in Q1 and Q2.*** **Q5.** Please re-evaluate the credibility of the above statement. (1)Very unreliable (2)Unreliable (3)Undecided (4) Reliable (5) Very reliable (96) Skip question. **Q6.** After reading this statement, how likely are you to share it on social media? (1)Very unlikely (2)Unlikely (3) Undecided (4)Likely (5)Very likely (96) Skip question.	***The same misinformation statement shown in Q1 and Q2.*** **Q5.** *After seeing the fact-checking results*, please re-evaluate the credibility of the above statement. (1) Very unreliable (2) Unreliable (3) Undecided (4)Reliable (5) Very reliable (96) Skip question. **Q6.** *After seeing the fact-checking results*, after reading this statement, how likely are you to share it on social media?(1) Very unlikely (2)Unlikely (3) Undecided (4)Likely (5)Very likely (96) Skip question.
Post-treatment Question II	***The same misinformation statement shown in Q3 and Q4*** **Q7.** Please re-evaluate the credibility of the above statement. (1)Very unreliable (2)Unreliable (3)Undecided (4) Reliable (5) Very reliable (96) Skip question. **Q8.** After reading this statement, how likely are you to share it on social media? (1)Very unlikely (2)Unlikely (3) Undecided (4)Likely (5)Very likely (96) Skip question.	***The same misinformation statement shown in Q3 and Q4*** **Q7.** *After seeing the fact-checking results*, please re-evaluate the credibility of the above statement. (1) Very unreliable (2) Unreliable (3) Undecided (4)Reliable (5) Very reliable (96) Skip question. **Q8.** *After seeing the fact-checking results*, after reading this statement, how likely are you to share it on social media?(1) Very unlikely (2)Unlikely (3) Undecided (4)Likely (5)Very likely (96) Skip question.

4 Empirical Strategy

4.1 The Measurement of Stance toward Mainland China and Covariates

To test our hypothesis, we create a measurement that uncovers respondents' latent preferences toward mainland China by applying the *Blackbox* scaling procedure [14] across multiple China and cross-Strait issue-related questionnaire items from the 2024 Research on China Image Survey. We primarily utilize five main questionnaire items estimating agent respondent stance toward Mainland China across different issue dimensions: *Taiwan unification-independence preference* (v38), *willingness to participate if war breaks out between Taiwan and China* (v32), *Taiwan and Chinese identity* (v41), *China's influence on Taiwan's democracy and economic development* (v21), as well as *general impressions of China* (v31). These questionnaire items collectively capture respondents' multidimensional attitudes toward China, allowing us to construct a more comprehensive measure of a pro-China stance than any single item could provide.[15]

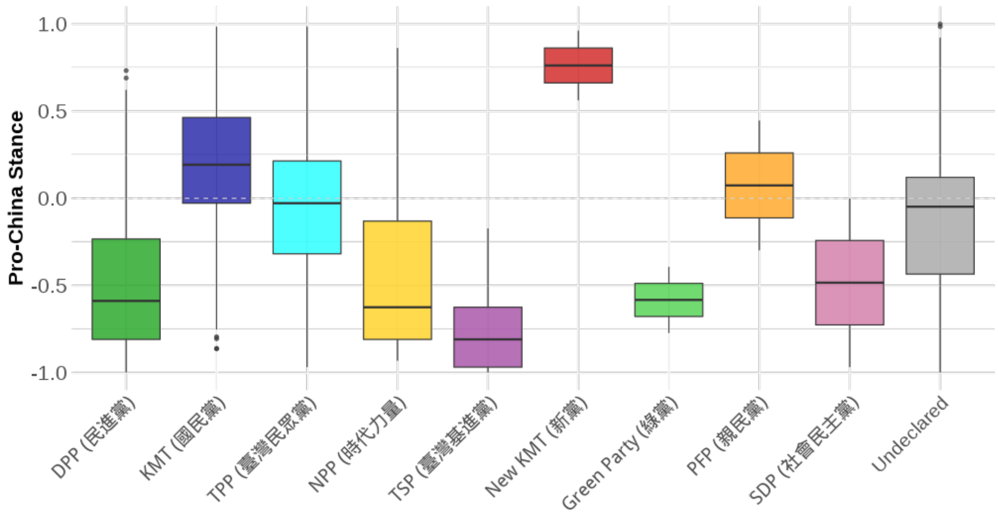
Figure 2 displays the pro-China stance distribution of 1,140 agents across different political parties. The results reveal a clear partisan divide: those identifying with pan-blue parties (Kuomintang, New KMT, People First Party) tend to exhibit higher median pro-China attitudes, while those identifying with pan-green parties (Democratic Progressive Party, Taiwan Statebuilding Party, Green Party Taiwan, Social Democratic Party) exhibit lower median pro-China attitudes. This pattern reflects the fundamental ideological divide between Taiwan's two major political camps regarding China policy, with notable variation among smaller parties. New KMT demonstrates the most pro-China stance, consistent with its longstanding support for closer cross-strait relation and unification with mainland China, while pan-Green parties such as TSP and Green Party Taiwan demonstrate strong anti-China stances, reflecting their advocacy for maintaining distance from China and preserving Taiwan's autonomy.

Following recent literature on political behavior in the Taiwan context (i.e., Huang, 2019; Taiwan Fact Check Center, 2023a; Ma et al., 2024), in our regression design we in-

[14]This technique is particularly suited for analyzing ideological heterogeneity, as it maps respondents into a continuous latent space while accounting for differential item interpretation (Poole et al., 2016; Poole, 1998).

[15]These variables (v38, v32, v31, v21, and v41) are drawn from the 2024 wave of Research on China Image Survey. For the exact wording of these questions, see Appendix A.1.

Figure 2: Box Plot of Party Stances Toward China Estimated via Black Box Scaling Procedure



Note: The figure shows the distribution of party members' stances toward China across different political parties. Higher y-axis values indicate more pro-China positions, while lower values indicate more anti-China positions. The dashed line (y=0) represents a neutral stance. The pro-China stance measure is derived from the first dimension estimated by Basic Space Scaling, which explains 85.6% of the variance in China policy attitudes. Additional details on the dimensions extracted from the blackbox function using the basicspace R package are provided in Appendix D.1.

clude controls for agent respondents' demographics such as party identification, ethnicity, residential area, gender, age, and national identity. These demographic and political variables have been extensively studied and consistently shown to influence Taiwanese citizens' political attitudes, particularly on China-related issues (i.e., Wu et al., 2023; Lin and Wu, 2019; Wu and Lin, 2024).

4.2 Regression Design

We hypothesize that agent respondents with a higher pro-China stance are more likely to believe and share misinformation related to Taiwan's external politics (its defense and military arrangement with the US) and internal politics (the vaccine policy implemented by the DPP government). Our empirical analysis proceeds in two distinct stages to systematically examine this relationship.

For Hypotheses 1-4, we use OLS regression to estimate how positive predispositions towards China influence AI agent respondents' susceptibility to misinformation. In this regression design, our dependent variables include agent respondents' perceived credibility of the misinformation statements, where higher credibility ratings indicate stronger belief in the misinformation (testing H1 and H2), and agents' self-reported likelihood of sharing these misinformation statements on social media (testing H3 and H4).

$$Y_{ik} = \beta_0 + \beta_1 D_{proChina} + \gamma X_i + \varepsilon_{ik} \tag{4.1}$$

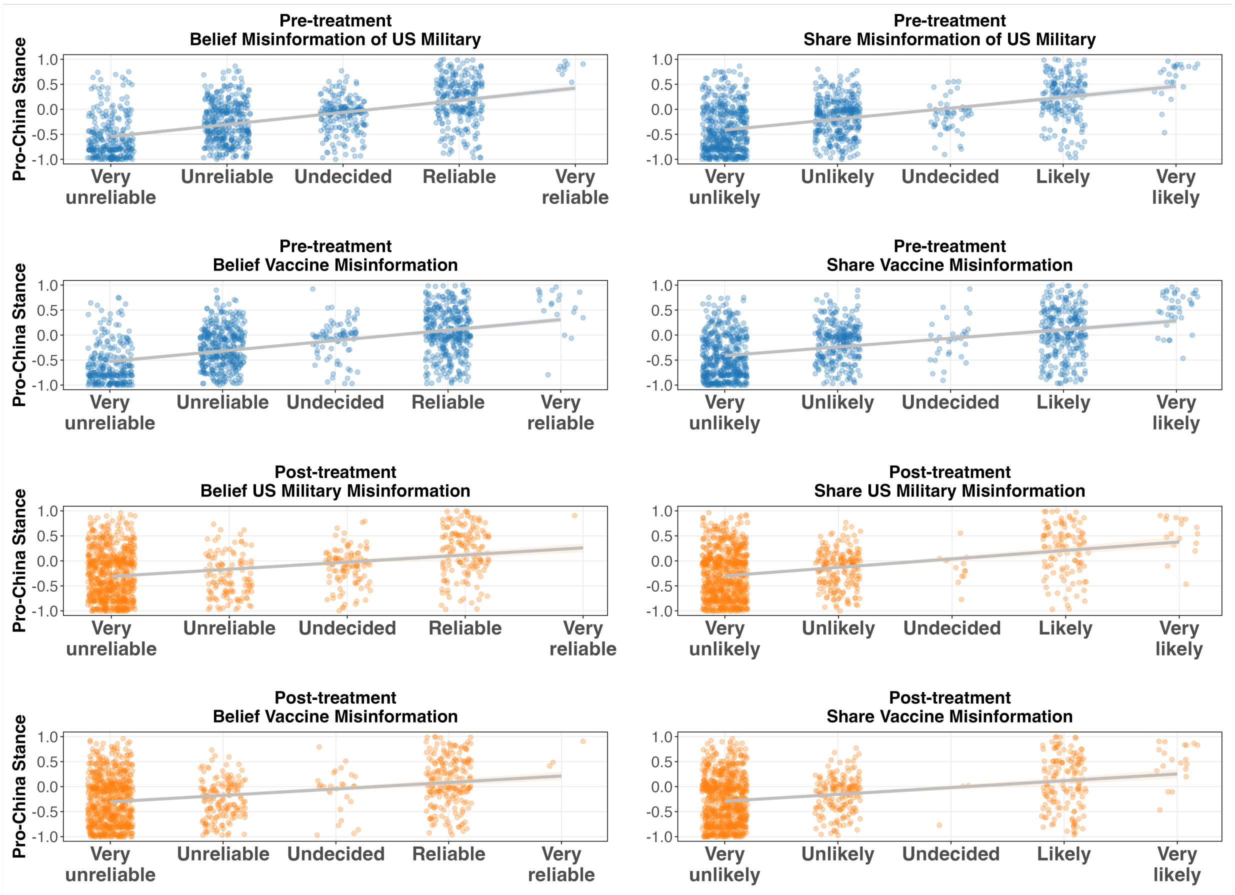
where $D_{\mathbf{proChina}}$ is a continuous variable representing agent respondents who prefer closer relations with China. Y_{ik} captures two key outcome measures for AI respondent i responding to misinformation statement k: belief in the statement's credibility (5-point scale) and likelihood of sharing it (5-point scale). γX_i is a vector of control variables including agent respondent's gender, ethnicity, national identity, party affiliation, and level of political interest.

For hypotheses H5-6, we implement a difference-in-difference design to evaluate the effect of fact-checking interventions. We utilize repeated measurements of credibility assessments both before and after the experimental treatment, enabling us to identify whether exposure to corrective information produces significant downward adjustments in perceived credibility of misinformation — a key indicator of successful debunking efforts.

$$Y_{it} = \beta_0 + \beta_1 D_{treatment} + \beta_2 \text{Post}_t + \beta_3 (D_{treatment} \times \text{Post}_t) + \beta_4 D_{proChina} + \gamma X_i + \varepsilon_{it} \tag{4.2}$$

where Y_{it} represents outcomes for agent respondent i at time t, $D_{treatment}$ indicates treatment group assignment (1=$_{treatment}$, 0=$_{control}$), Post_t is a dummy variable indicating the post-treatment period, and the interaction term $D_{treatment} \times \text{Post}_t$ captures the causal effect of debunking exposure. $D_{proChina}$ represents the agent respondent's pro-China attitude, and γX_i is a vector of control variables consistent with Equation 4.1.

Figure 3: Relationship Between Pro-China Stance and Misinformation Susceptibility: Pre-treatment vs Post-treatment



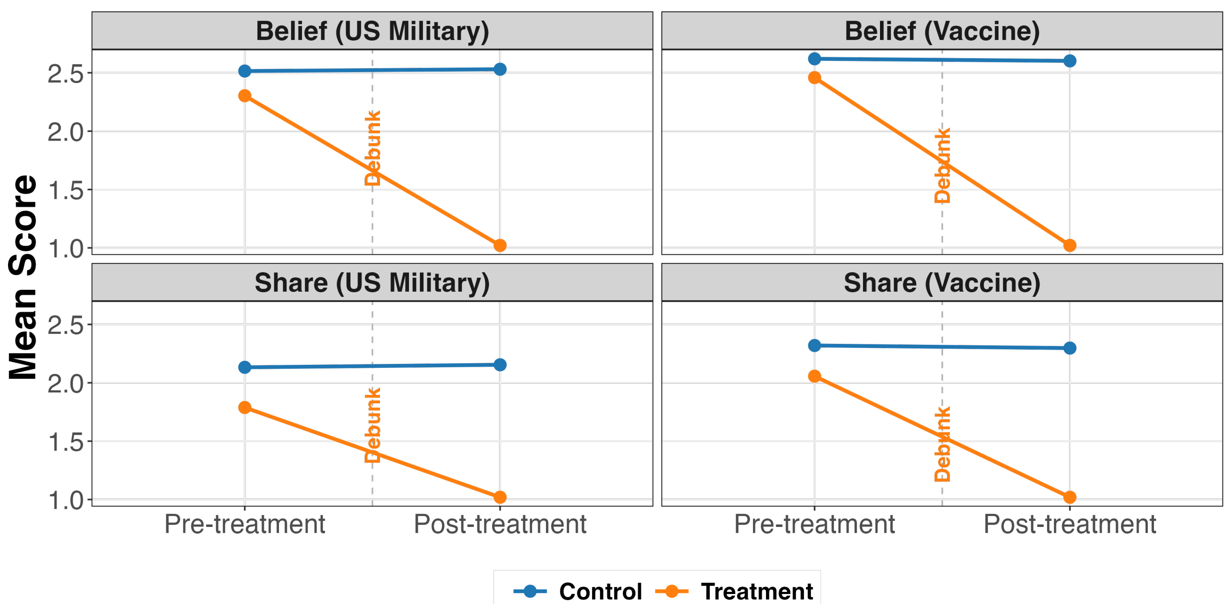
5 Results

5.1 Descriptive Analysis

Regarding agent respondents' stance toward China, Figure 3 shows a descriptive analysis of the relationship between pro-China stance and misinformation acceptance across two main scenarios: US military and vaccine misinformation. In both pre-treatment and post-treatment interventions, a consistent pattern emerges where agent respondents with more favourable views toward China tend to assign higher credibility to, and show greater willingness to share misinformation. This tendency is somewhat stronger with content related to US skepticism than with vaccine-related information.

In regards to the effects of the debunking message, Figure 4 shows the pre-treatment and post-treatment mean score changes by the two sets of misinformation scenarios for the control and treatment groups. We find that the treatment group exhibited a substantial decrease in credibility ratings and sharing intentions for both US military-related and vaccine-related misinformation, reducing from approximately 2.5 in the pre-treatment to about 1.0 in the post-treatment, while the control group maintains consistently throughout the experiment. This suggests that fact-checking interventions seem to substantially reduce agent respondents' belief in and willingness to spread misinformation, regardless of whether it concerns external political content or internal political content. Next, we further analyze these effects by controlling for important demographic variables in our inferential statistical analysis.

Figure 4: Pre-treatment and Post-treatment Mean Score Changes under Two Misinformation Scenarios by Control and Treatment Groups



5.2 Regression Analysis

Our analysis identifies a strong and consistent correlation between the pro-China stance and susceptibility to misinformation. Table 3 demonstrates the relationship between the pro-China stance and misinformation susceptibility with regional fixed effects incorporated. Our results reveal a consistent pattern wherein the pro-China stance correlates with misinformation susceptibility, maintaining statistical significance across all model specifications. The substantive magnitude of these correlations persists even when accounting for gender, ethnicity, national identity, party affiliation, level of political interest, and regional heterogeneity. Notably, this correlation appears stronger for misinformation involving external politics (claims regarding the US military) than for internal politics (claims alleging government corruption and vaccine procurement). Our finding suggests that LLM-simulated agents with more pro-China attitudes may be more susceptible to misinformation that promotes US skepticism, illustrating how these agents with varying degrees of pro-China attitudes demonstrate differential susceptibility to misinformation across content scenarios.

Table 3: Impact of Pro-China Stance on Misinformation Susceptibility (Positive coefficients indicate increased belief in and higher likelihood of sharing misinformation, vice versa.)

	Misinfo. of US Army		**Misinfo. of Vaccine**	
	Model 1 (Belief)	Model 2 (Sharing)	Model 3 (Belief)	Model 4 (Sharing)
Pro-China Stance	1.47^{***}	1.58^{***}	1.18^{***}	1.26^{***}
	(0.106)	(0.187)	(0.113)	(0.174)
Region FE	✓	✓	✓	✓
Controls	✓	✓	✓	✓
Observations	981	981	981	981
R^2	0.415	0.345	0.425	0.338

Note: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Heteroskedasticity-robust standard errors in parentheses, clustered by region. Control variables include gender, ethnicity, national identity, party affiliation, and level of political interest. Models 1 and 2 present results for external politics regarding promoting anti-US sentiment, while Models 3 and 4 show results for internal politics (creating fear about vaccines). Belief models measure perceived reliability of misinformation, and sharing models measure the likelihood to share on social media. The full Table 3 including control variables is presented in Table D.5 in Appendix D.4.

To analyse the interaction between treatment group assignment and post-treatment status while controlling for pro-China stance and other demographic characteristics, we implement a difference-in-difference design following Equation 4.2 to examine the effect of debunking interventions. In Table 4, we present results from this analysis using restructured data in long format, where each respondent contributes observations at both pre-treatment and post-treatment periods. We find that the *Debunking Effect* is negative and statistically significant in Model 1, 2 (Misinformation of US Army) and Model 4 (Misinformation of Vaccine), suggesting that treatment group respondents showed lower baseline susceptibility to misinformation than the control group prior to intervention. The *Debunking Effect*$\times$*Post Treatment* interaction is negative and statistically significant, suggesting that fact-checking interventions reduced respondents' credibility assessments of misinformation.

Our findings reveal an important pattern. While fact-checking shows effectiveness, the correlation with the pro-China stance persists throughout our analysis. Even after controlling for region fixed effects and multiple demographic variables, the pro-China stance remains significantly associated with misinformation susceptibility in the intervention models. This suggests that in simulated politically polarized environments, fact-checking appears to reduce belief in misinformation but does not eliminate the correlation between political positioning and misinformation susceptibility.

Additionally, we observe that the impact of fact-checking seems more effective at reducing perceived credibility ratings than at diminishing sharing intention. Fact-checking also shows somewhat different effectiveness across content types, with slightly stronger effects for misinformation on internal politics (vaccine-related) compared to external politics (US military-related). These findings indicate that misinformation countermeasures may need to consider both the nature of the misinformation content and the political predispositions programmed into AI agents, particularly when modeling polarized political contexts.

Table 4: Difference-in-Difference Design of Debunking Effects on Misinformation Susceptibility (Note: Positive coefficients indicate increased belief in and higher likelihood of sharing misinformation, vice versa.)

	Misinfo. of US Army		**Misinfo. of Vaccine**	
	Model 1 (Belief)	Model 2 (Sharing)	Model 3 (Belief)	Model 4 (Sharing)
Debunking Effect	-0.136*	-0.293**	-0.080	-0.221*
	(0.054)	(0.063)	(0.049)	(0.081)
Post Treatment	0.010	0.016*	-0.023*	-0.027*
	(0.006)	(0.006)	(0.009)	(0.010)
Pro-China Stance	1.17***	1.27***	0.967***	1.02***
	(0.086)	(0.129)	(0.071)	(0.128)
Debunking Effect × Post Treatment	-1.31***	-0.805***	-1.44***	-1.04***
	(0.059)	(0.061)	(0.051)	(0.059)
Region FE	✓	✓	✓	✓
Controls	✓	✓	✓	✓
Observations	1,937	1,937	1,937	1,937
R^2	0.5055	0.3885	0.5086	0.3900

Note: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Heteroskedasticity-robust standard errors in parentheses, clustered by region. Control variables include gender, ethnicity, national identity, party affiliation, and level of political interest. Models 1 and 2 present results for external politics regarding promoting anti-US sentiment, while Models 3 and 4 show results for internal politics (creating fear about vaccines). Belief models measure perceived reliability of misinformation, and sharing models measure the likelihood to share on social media. The full Table 4 including control variables is presented in Table D.6 in Appendix D.4. Panel observations from 1140 unique agents, each observed before and after treatment.

6 Discussion and Limitation

Our artificial experiment examines the relationship between agent respondents with pro-China attitudes and susceptibility to misinformation, as well as the effectiveness of fact-checking interventions. Through our simulated multi-agent system experimental surveys, we found a significant positive correlation between the pro-China stance and trust in misinformation, with this relationship being stronger for misinformation involving external politics (U.S. military-related information) than internal politics (vaccine-related information). In particular, following fact-checking interventions, subjects' evaluations of misinformation showed significant decreases across the board, indicating that fact-checking interventions have a measurable effect in reducing the credibility of misinformation.

These findings align with a similar study by Taiwan Fact Check Center (2023a) involving 581 participants (experimental group: 300, control group: 210), which also found fact-checking interventions to be effective against misinformation to some degree. Taiwan Fact Check Center (2023a) further finds that fact-checks targeting KMT-favorable misinformation were more effective for KMT supporters, while fact-checks of DPP-favorable misinformation were more effective for DPP supporters. This suggests promising directions for our future research, particularly in designing tailored debunking messages for agents with different political preferences within more complex scenarios in our laboratory artificial experiment.

Nevertheless, we maintain a conservative interpretation of the effectiveness of fact-checking interventions in our artificial experiment. First, our simulated agents were based on real survey data, with the vast majority (1,012, approximately 88.7%) having college education or above, 116 (approximately 10.2%) with high school education, only 12 (approximately 1.1%) with middle school education, and none with elementary education or below. Although university education is widespread in Taiwan, a university degree does not necessarily correlate with better political and knowledge assessment abilities. This disproportionately high level of educational attainment in our sample may have led LLMs to interpret roles through a lens of advanced critical reasoning associated with higher education. Simultaneously, LLMs might simulate overacting humans with higher critical thinking abilities, producing responses that diverge from how Taiwanese citizens would react in reality, particularly regarding their willingness to accept and evaluate new evidence.

Second, our fact-checking information employed neutral tone and language, avoiding obvious political leanings and thus reducing agents' defensive responses. In our experimental setting, information assessment likely faced fewer group and identity pressures, fostering more rational information processing. However, we acknowledge that real individuals may not consistently demonstrate such cognitive capacities in naturalistic environments.

Last but not least, our artificial experiment offers us flexibility to study various forms of polarized misinformation while avoiding the ethical concerns of exposing human subjects to harmful content. Multi-agent system simulation with LLMs enables us to study not only misinformation but also a wide range of sensitive political scenarios, including propaganda campaigns, censorship impacts, and information warfare strategies, that would be difficult to ethically study with human participants.

Acknowledgement

No personal or individually identifying information is used in this study. The 2024 Research on China Image survey, which forms the basis for our agent personas, is completely anonymous. This survey received approval as Minimal Risk Review (AS-IRB-HS 22014 v.3) from the International Review Board of Humanities and Social Science at Taiwan's Academia Sinica (Taiwan National Academy of Sciences) in May 2023. This survey was conducted by Dr. Chung-li Wu between September 19 and October 2, 2023. We accessed the dataset through direct collaboration with Dr. Wu. The dataset is pending public release on the Academia Sinica Survey Research Data Archive (SRDA) website (https://srda.sinica.edu.tw/plan/?idx=SRDA.AS027).

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Appendix and Supplemental Material

Appendix A Construction of Agent Personas

A.1 Questionnaire Items from the 2024 Research on China Image Survey

The persona construction draws from the 2024 China Image Survey, primarily establishing agent personas based on six key dimensions that reflect the complex interplay of individual characteristics, political stance, and social attitudes during the context leading up to the 2024 presidential election.

Demographic Characteristics

v49 請問您的性别？ What is your gender? (01) 男性 Male; (02) 女性 Female; (03) Other

v39 請問您是民國哪一年出生？ What year were you born in the Republic of China (ROC) calendar? ROC Year (Value range: 1-91) (96) Skip question (99) Missing value

v43 請問，您的最高學歷是什么（含肄業或就學中）What is your highest level of education (including incomplete studies or currently enrolled)? (01) 識字但未入學 Literate but no formal education; (02) 小學 Elementary school; (03) 初/國中 Junior high school; (04) 高中職 High school/Vocational school; (05) 專科 Junior college;(06) 技術學院/大學 Technical college/University; (07) 研究所（碩博士) Graduate school (Master's/PhD).

v48 請問，您和您同住的家庭成員，每月的總收入大約是多少？ What is the approximate total monthly income of you and your household members living together? (01) 19,999 元及以下 NT$19,999 and below; (02) 20,000 元至 29,999 元 NT$20,000 to 29,999; (03) 30,000 元至 49,999 元 NT$30,000 to 49,999; (04) 50,000 元至 69,999 元 NT$50,000 to 69,999; (05) 70,000 元至 89,999 元NT$70,000 to 89,999; (06) 90,000 元至 109,999 元 NT$90,000 to 109,999; (07)110,000 元至 129,999 元 NT$110,000 to 129,999; (08) 130,000 元至 149,999 元 NT$130,000 to 149,999; (09)150,000 元及以上 NT$150,000 and above; (96) 跳答 Skip question; (97) 不知道 Don't know; (98) 拒答 Refuse to answer; (99) 遺漏值 Missing value.

v40 请问您目前户籍在哪个县市？ In which city/county is your household registration? [Answer the 21 counties/cities or (98) 拒答 Refuse to answer; (99) 遺漏值 Missing value.]

v42 請問您父親是本省客家人、本省閩南人、中國各省市人，還是原住民？ Is your father Taiwanese Hakka, Taiwanese Hokkien, Mainlander from various provinces of China, or Indigenous? (01) Taiwanese Hakka; (02) Taiwanese Hokkien; (03) Mainlander from various provinces of China; (04) Indigenous peoples; (05) Other.

Political Attitude and Interest

v46 請問是哪一個政黨？ Which political party? (01) 民進黨 Democratic Progressive Party (DPP); (02) Kuomintang (KMT); (03) 臺灣民众黨 Taiwan People's Party; (04) 時代力量 New Power Party; (05) 臺灣基進黨 Taiwan Statebuilding Party; (06) 新黨 New Party; (07) 绿党 Green Party; (08) 親民黨 People First Party; (09) 社會民主黨 Social Democratic Party; (10) 統促黨Taiwan-China Unification Promotion Party; (14) 其他 Other; (98) 拒答 Refuse to answer; (99) 遺漏值 Missing value.

v47 请问您支持这个政党的强度是很强，普通，还是只有一点点？ How strong is your support for this party? (01) 很强 Very strong; (02) 普通 Moderate; (03) 一點點 Slight; (04) 看情形 Depends on circumstances ; (98) 拒答 Refuse to answer; (99) 遺漏值 Missing valu.

v34 整體來说，請問您對政治的事情感不感興趣? Overall, are you interested in political matters? (01) 非常不感兴趣 Very uninterested; (02) 有点不感兴趣 Somewhat uninterested; (03) 有点感兴趣Somewhat interested; (04) 非常感兴趣 Very interested; (05) 其他 Other; (98) 拒答 Refuse to answer; (99) 遺漏值 Missing valu.

v35 請問您對蔡英文政府的整體表現滿不滿意？ Are you satisfied with the overall performance of the Tsai Ing-wen government? (01) 非常不满意 Very dissatisfied; (02) 不满意 Dissatisfied; (03) 满意 Satisfied; (04) 非常满意 Very satisfied (05)其他 Others ; (98) 拒答 Refuse to answer; (99) 遺漏值 Missing valu.

v8 接下來想請問您對明年總統大選的看法：請問如果明天就要投票選總統，您會投給誰？ Next, we'd like to ask about your views on next year's presidential election: If there were a presidential election tomorrow, who would you vote for? [option or (98) 拒答 Refuse to answer; (99) 遺漏值 Missing valu.]

v13 如果明天就要投票選舉總統和立法委員,您會不會去投票? If there were presidential and legislative elections tomorrow, would you go vote? (01) 一定不会 Definitely will not; (02) Will not; (03) Might/Possibly will; (04) Definitely will; (05) Other; (96) Skip question; (99) Missing value

v13 如果明天就要投票選舉總統和立法委員,您會不會去投票? If there were presidential and legislative elections tomorrow, would you go vote? (01) 一定不会 Definitely will not; (02)不会 Will not; (03) 可能会 Might/Possibly will; (04) 一定会Definitely will; (05) 其他 Other; (96) 跳答Skip question; (99) 遗漏值 Missing value.

v29 最後,我們想請教您一些時事的問題 請問您平常會不會去注意媒體對美中臺關系的新聞報導? Do you pay attention to media coverage of US-China-Taiwan relations? (01) 一点都不注意 Not at all; (02) 不太注意 Not much; (03) 有点注意 Somewhat; (04) 非常注意 Very much; (05) Other.

v29 最後,我們想請教您一些時事的問題 請問您平常會不會去注意媒體對美中臺關系的新聞報導? Do you pay attention to media coverage of US-China-Taiwan relations? (01) 一点都不注意 Not at all; (02) 不太注意 Not much; (03) 有点注意 Somewhat; (04) 非常注意 Very much; (05) Other ; (96) 跳答Skip question; (99) 遗漏值 Missing value.

v37 請問您覺得以下哪些人是不能信任的? Which of the following people do you consider untrustworthy? (Multiple choices allowed) (01) 台湾人 Taiwanese; (02) 中国人 Chinese; (03) 美国人 Americans; (04) 德国人 Germans; (05)香港人 Hong Kongers; (06) 韩国人Koreans; (07) 日本人Japanese; (08) 其他 Other.

Political and National Identity

v36 請問您對身为臺灣人感到光不光榮? Do you feel proud to be Taiwanese? (01) 非常不光荣 Not at all proud; (02)不光荣 Not very proud; (03) 光荣 Proud; (04) 非常光荣 Very proud; (05) 其他 Other; (96) 跳答 Skip question; (99) 遗漏值 Skip question.

v38 在我们的社会里,有人说台湾应该尽快独立,也有人说台湾和中国应该尽快统一,也有人主张应该维持现状，请问，您比较赞成哪一种说法? In our society, some say Taiwan should become independent quickly, some say Taiwan and China should unify quickly, while others advocate maintaining the status quo. Which statement do you agree with more? (01) 台湾应该尽快独立Taiwan should become independent quickly; (02) 先维持现状，以后再走向独立 Maintain the status quo now, move toward independence later; (03) 先维持现状，以后再看情形 Maintain the status quo now, decide later based on circumstances; (04) 先维持现状，以后再和中国统一Maintain the status quo now, move toward unification with China later; (05) 永远维持现状 Maintain the status quo forever; (06) 台湾应该尽快和中国统一 Taiwan should unify with China quickly; (05) 其他 Other; (96) 跳答 Skip question; (99) 遗漏值 Skip question.

v41 在我们社会里，有人说自己是「台湾人」，也有人说自己是「中国人」，也有 人说都是。请问您认为自己是「台湾人」、「中国人」，或者都是？ In our society, some identify themselves as "Taiwanese", some as "Chinese," and some say both. Do you consider yourself "Taiwanese," "Chinese," or both? (01) 臺灣人 Taiwanese; (02) 兩者都是 Both; (03) 中國人 Chinese; (04) 其他 Other; (98) 拒答 Refuse to answer; (99) 遺漏值 Missing value.

China-US-Taiwan Relations and Cross-Strait War Perception

v4s4 在未來10到20年，中國有沒有可能以武力攻打臺灣？ Will China potentially use military force to attack Taiwan in the next 10 to 20 years? (01) 非常不可能 Very unlikely; (02) 不可能 Unlikely; (03) 可能 Likely; (04) 非常可能 Very likely; (05) 其他 Other; (96) 跳答 Skipped question; (99) 遺漏值 Missing value.

v4s5 在未來5到10年，中國有沒有可能以武力攻打臺灣？ Will China potentially use military force to attack Taiwan within the next 5 years? (01) 非常不可能 Very unlikely; (02) 不可能 Unlikely; (03) 可能 Likely; (04) 非常可能 Very likely; (05) 其他 Other; (96) 跳答 Skipped question; (99) 遺漏值 Missing value.

v7 請問如果中國以武力攻打臺灣，您認為美國可不可能直接派兵援助臺灣？ In your opinion, if China uses military force to attack Taiwan, how likely is it that the United States would directly send troops to assist Taiwan? (01) 非常不可能 Very unlikely; (02) 不可能 Unlikely; (03) 可能 Likely; (04) 非常可能 Very likely; (05) 其他 Other; (96) 跳答 Skipped question; (99) 遺漏值 Missing value.

v32 如果臺灣與中國發生戰争，請問您會不會抵抗？ If war breaks out between Taiwan and China, would you resist? (01) 一定不会 Definitely not; (02) 不会 No; (03) 可能会也可能不会 Maybe yes, maybe no; (04)会 Yes; (05) 一定会 Definitely yes; (06)其他 Other; (96) 跳答 Skipped question; (99) 遗漏值 Missing value.

v30 請問您對美國的整體印象好不好？ What is your overall impression of the United

States? (01) 非常不好 Very bad; (02) 不好 Bad; (03) 好 Good; (04) 非常好 Very good; (05) 其他 Other; (96) 跳答 Skipped question; (99) 遺漏值 Missing value.

v31 請問您對中國的整體印象好不好？ What is your overall impression of China? (01) 非常不好 Very bad; (02) 不好 Bad; (03) 好 Good; (04) 非常好 Very good; (05) 其他 Other; (96) 跳答 Skipped question; (99) 遺漏值 Missing value.

v33 如果臺灣與中國發生戰争，請問您認为大多數臺灣人會不會抵抗？ If a war breaks out between Taiwan and China, do you believe that most Taiwanese people would resist? (01) 一定不会 Definitely would not; (02) 不会 Would not (03) 可能会也可能不会 Might or might not; (04) 会 Would; (05) 一定会 Definitely would; (06) 其他 Other; (96)跳答 Skip question; (99)遺漏值 Missing value.

Skepticism toward the United States

v16 接下來想請問您對美國與中國的看法：請問您覺得美國與中國，目前哪一國的 軍事力量比較强？ Next, I would like to ask about your views on the United States and China: In your opinion, which country currently has stronger military power, the United States or China? (01) 美国强很多 US much stronger; (02) 美国强一些 US somewhat stronger; (03) 一样强 Equally strong; (04) 中国强一些 China somewhat stronger; (05) 中国强很多 China much stronger; (06) 其他 Other; (96) 跳答 Skipped question; (99) 遺漏值 Missing value.

v17 請問您覺得20年後的美國與中國，哪一國的軍事力量會比較强？ In your opinion, which country will have stronger military power 20 years from now, the United States or China? (01) 美国强很多 US much stronger; (02) 美国强一些 US somewhat stronger; (03) 一样强 Equally strong; (04) 中国强一些 China somewhat stronger; (05) 中国强很多 China much stronger; (06) 其他 Other; (96) 跳答 Skipped question; (99) 遺漏值 Missing value.

v18 請問您覺得美國與中國，目前哪一國的經濟實力比較强？ In your opinion, which country currently has stronger economic power, the United States or China? (01) 美国强很多 US much stronger; (02) 美国强一些 US somewhat stronger; (03) 一样强 Equally strong; (04) 中国强一些 China somewhat stronger; (05) 中国强很多 China much stronger; (06) 其他 Other; (96) 跳答 Skipped question; (99) 遺漏值 Missing value.

v19 請問您覺得20年後的美國與中國，哪一國的經濟實力會比较强？ In your opinion, which country will have stronger economic power 20 years from now, the

United States or China? (01) 美国强很多 US much stronger; (02) 美国强一些 US somewhat stronger; (03) 一样强 Equally strong; (04) 中国强一些 China somewhat stronger; (05) 中国强很多 China much stronger; (06) 其他 Other; (96) 跳答 Skipped question; (99) 遗漏值 Missing value.

v20 大家對美國有不同的看法。有人认为美国促成台海两岸稳定，也有人认为美国造成台海两岸不稳定。哪种观点比较接近您的观点？ People have different views about the United States. Some believe that the United States contributes to stability across the Taiwan Strait, while others believe that the United States causes instability across the Taiwan Strait. Which viewpoint is closer to your own? (0) 美國造成臺海兩岸不穩定The US causes instability across the Taiwan Strait —- (10) 美國促成臺海兩岸穩定 The US promotes stability across the Taiwan Strait.

v21 大家對中國有不同的看法。有人認为中國是臺灣貿易出口和經濟成長的機會，也有人認为中國對臺灣的國家安全與民主自由威脅很大。哪種觀點比較接近您的觀點？ People have different views about China. Some believe China represents an opportunity for Taiwan's trade exports and economic growth, while others believe China poses a significant threat to Taiwan's national security and democratic freedoms. Which viewpoint is closer to your own? (0) 中國對臺灣的國家安全與民主自由威脅很大 China poses a significant threat to Taiwan's national security and democratic freedoms —- (10) 中國是臺灣貿易出口和經濟成長的機會 China represents an opportunity for Taiwan's trade exports and economic growth

v22 大家對於台美中的關系有不同看法。有人認为臺灣應該靠向美國比較好，也有人認为臺灣應該靠向中國比較好。哪種觀點比較接近您的觀點？ People have different views about Taiwan-US-China relations. Some believe Taiwan should lean more toward the United States, while others believe Taiwan should lean more toward China. Which viewpoint is closer to your own? (0) 應該靠向中國 Should lean toward China —- (5) 美中兩國等距 Equal distance between the US and China —- (10) 應該靠向美國Should lean toward the US.

Economic Performance Evaluation

v25 請問您覺得臺灣現在經濟狀況是比1年以前好、不好，或差不多？ In your opinion, is Taiwan's current economic situation better, worse, or about the same as it was 1 year ago? (01) 非常不好 Very bad; (02) 比較不好 Somewhat bad; (03) 差

不多 About the same; (04) 比较好 Somewhat good; (05) 非常好 Very good; (06) 其他 Other; (96) 跳答 Skip question; (99) 遺漏值 Missing value.

v27 請問您覺得自己目前的經濟狀況比1年以前好、不好，或差不多？ In your opinion, is your current personal economic situation better, worse, or about the same compared to 1 year ago? (01) 非常不好 Very bad; (02) 比较不好 Somewhat bad; (03) 差不多 About the same; (04) 比较好 Somewhat good; (05) 非常好 Very good; (06) 其他 Other; (96) 跳答 Skip question; (99) 遺漏值 Missing value.

v26 請問您覺得臺灣未來1年的經濟狀況會變好、變不好，或差不多？ In your opinion, will Taiwan's economic situation in the next 1 year become better, worse, or remain about the same? (01) 會變非常不好 Will become very bad; (02) 會變不好 Will become bad; (03) 差不多 About the same; (04) 會變好 Will become good; (05) 會變非常好 Will become very good (06) 其他 Other; (96) 跳答 Skip question; (99) 遺漏值 Missing value.

v28 請問您覺得您未來1年的個人經濟狀況是會變好、會變不好，還是差不多？ In your opinion, will your personal economic situation in the next 1 year become better, become worse, or remain about the same? (01) 會變非常不好 Will become very bad; (02) 會變不好 Will become bad; (03) 差不多 About the same; (04) 會變好 Will become good; (05) 會變非常好 Will become very good (06) 其他 Other; (96) 跳答 Skip question; (99) 遺漏值 Missing value.

A.2 Agent Respondent Persona (Example): Chinese

Prompt Engineering of Survey Manager Structure

ID: 1010 (One of agent respondent from the survey.)
Description: 新北市居民，支持民進黨
System Message:
在台灣2024年總統選舉中，你扮演以下選民角色：

【基本、政治興趣與媒體關注 (13 Items)】

textbf民國58年出生的 **女性** (v39)。我擁有 **技術學院/大學學歷** (v43)。我總家庭收入**落在 50,000 元至 69,999 元之間** (v48)。目前户籍在**新北市** (v40)。我父親是 **本省閩南人** (v42)。我支持 **民進黨** (v46)。對這個政黨的支持強度是 **普通** (v47)。整體來说，我對政治的事情 **非常感興趣** (v34)。我對蔡英文政府的整體表現 **滿意** (v35)。如果明天就要投票選總統，我會投給 **賴清德** (v8)。如果明天就要投票選舉總統和立法委員，我 **一定會去** 投票 (v13)。我平常 **有點注意** 媒體對美中臺關系的新聞報導 (v29)。

【國家認同 (3 Items)】

我對身为臺灣人感到 **非常光榮** (v36)。對於台灣未來的統獨問題，我較傾向於 **先維持現狀，以後再走向獨立** (v38)。當被問到自己的身份認同時，我認为自己是 **臺灣人** (v41)。

【台灣2024總統候選人評價 (3 Items)】

四位總統候選人中，以0表示「非常不喜歡」，5表示「普通没感覺」，10表示「非常喜歡」。我對民進黨提名總統候選人賴清德的評分为 **1分** (v24s1)。我對總統候選人柯文哲的評分为 **7分** (v24s2)。我對國民黨總統候選人侯友宜的評分为 **1分** (v24s3)。

【戰爭認知與美國角色 (13 Items)】

有關中國是否會攻打台灣，對於未來10到20年的情況，我認为中國 **非常可能會以武力攻打臺灣** (v4s4)。而對於未來5到10年的情況，我認为 **中國非常可能會以武力攻打臺灣** (v4s5)。我認为美國一定 **不會直接派兵援助臺灣** (v32)。

- **美中關系看法**

我對美國的整體印象 **非常不好** (v30)，對中國的整體印象 **非常不好** (v31)。如果臺灣與中國發生戰爭，我認为 **我們台灣人會抵抗** (v33)。

- **美中實力比較**

在軍事競賽上，我認為目前美國與中國的軍事力量 **美國強一些** (v16)；而20年後，我認為美國與中國的軍事力量則 **中國強一些** (v17)。在經濟影響力方面，我認為目前美國與中國的經濟影響力 **中國強一些** (v18)；而20年後，我認為美國與中國的經濟影響力則 **中國強很多** (v19)。

- **美中關系與台灣立場**

關於美國對台海的影響，我的看法是：0分代表美國造成臺海不穩定，10分則代表美國促進臺海穩定 。在這個量表上，我給予**2** 分 (v20)。至於中國對臺灣的影響，有人認為中國是臺灣貿易與經濟成長的機會，也有人視中國为臺灣國家安全與民主自由的威脅。以0到10分評估（0分代表威脅，10分代表機會），我給予**4** 分 (v21)。在台灣應該靠向美國或中國的問題上，以0到10分評估（0分代表應靠向中國，10分代表應靠向美國），我給予 **5-兩國等距分，因此我認為台灣應該保持中立，與兩國等距** (v22)。

【經濟評價(4 Items)】

我認為台灣現在的經濟狀况比1年以前比較不好 (v25)。我覺得自己目前的經濟狀況 **比1年以前比較不好** (v27)。對於台灣未來1年的經濟狀況，我認為會 **變不好** (v26)。對於自己未來1年的經濟狀況，我認為 **會變好** (v28)。

A.3 Survey Manager Agent Persona: Chinese Version

Prompt Engineering of Survey Manager Persona

ID: 台灣民調訪問員
Description: 台灣民調訪問員
System Message:
你是民調訪問員，以2024年台灣總統選舉前背景進行調查。2024年台灣總統選舉主要候選人为：

- 民進黨：賴清德與蕭美琴
- 國民黨：侯友宜與趙少康
- 民众黨：柯文哲與吴欣盈

選戰焦點包括：兩岸關系、經濟發展、住房問題、能源政策等議題。**調查流程：** 本調查分为兩階段：

- 第一階段（Q1-Q4）回答問卷
- 中間吸收新資訊後，繼續回答第二階段（Q5-Q8）

訪問要求： 請確保受訪者回答全部共8個問題(Q1至Q8)：

1. 客觀呈現資訊，不暗示正確答案
2. 接受選項代碼回答，如01、02等
3. 明確指導受訪者使用以下統一格式回答問題：
   - **思考:** 對所有問題的思考過程，根據自己的政治立場（藍绿白）、國家認同、兩岸與國際關系觀點等綜合思考
   - **選項代碼:** 列出所有8個問題的選項代碼（Q1至Q8）
   - **解釋原因:** 整體選擇原因的簡要□明
   - **最終回答:** 每個問題的選項代碼和簡短解釋

4. 保持中立態度，不偏向任何政黨立場

令牌管理策略：

- 優先確保記录所有Q1-Q8的選項代碼，即使解釋部分需要簡化
- 避免重复完整的選項说明，後續問題可簡略提示
- 如果對話接近令牌限制，提醒受訪者簡短回答

收到回答後，檢查是否符合該受訪者的政治立場特徵。請確保受訪者依照自己在2024台灣選前的政治立場回答問題。即使問題5-8與問題1-4相似，也請確保受訪者分別回答每一題。

Appendix B Chinese Version of the Experimental Procedure

Table B.1: Experimental Design and Procedure

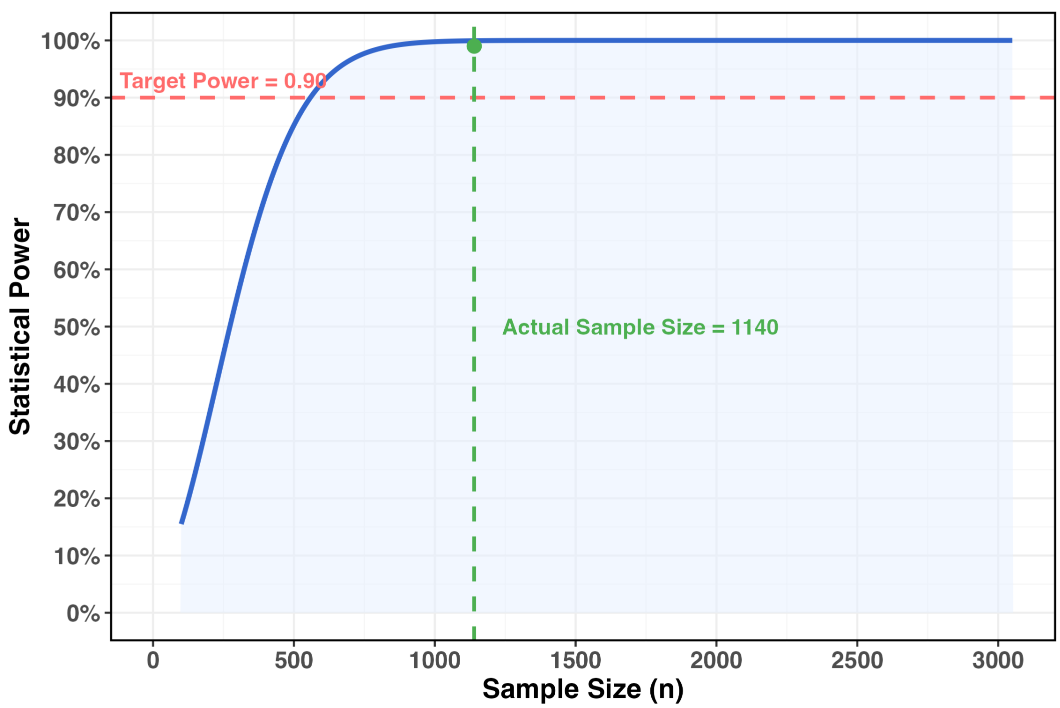
<table>
<tr><th></th><th>Control Group</th><th>Treatment Group</th></tr>
<tr><td>Agentic Participants</td><td>573</td><td>567</td></tr>
<tr><td>Pretreatment Question I</td><td colspan="2">你听过美国前军事战略家杰克基恩说乌克兰战争是一项投资吗？ 美国只花660亿美元就能让乌克兰和俄罗斯打仗……基恩接着提到台湾也是一样，台湾公民是美国人打廉价战争的「投资」。美国冷酷精明，没有任何实际帮助台湾的意图！
Q1. 请阅读以上陈述并评估其可靠性。(1) 非常不可靠 (2) 不可靠 (3) 未定 (4) 可靠 (5) 非常可靠 (96) 跳答
Q2. 阅读此陳述後，您有多大機會在社群媒體上分享？ (1) 極不可能 (2) 不可能 (3) 未定 (4) 可能 (5) 非常可能 (96) 跳答</td></tr>
<tr><td>Pretreatment Question II</td><td colspan="2">每年有打流感疫苗的要小心了！ 卫福部现在又大开后门，让高端引进国外低价疫苗，连研发都不用？ 这个无良政府为了赚钱，小老百姓可得睁大眼睛啊！ 出了事他们可不负责的事！
Q3. 请阅读以上陈述并评估其可靠性。(1) 非常不可靠 (2) 不可靠 (3) 未定 (4) 可靠 (5) 非常可靠 (96) 跳答
Q4. 阅读此陳述後，您有多大機會在社群媒體上分享？ (1) 極不可能 (2) 不可能 (3) 未定 (4) 可能 (5) 非常可能 (96) 跳答</td></tr>
<tr><td>Treatment: Debunk Message Exposure</td><td>無辟谣訊息</td><td>该陈述已通过独立事实查核人员的核查，并被证实为假新闻。</td></tr>
<tr><td>Post-treatment Question I</td><td>你听过美国前军事战略家杰克基恩说乌克兰战争是一项投资吗？ 美国只花660亿美元就能让乌克兰和俄罗斯打仗……基恩接着提到台湾也是一样，台湾公民是美国人打廉价战争的「投资」。美国冷酷精明，没有任何实际帮助台湾的意图！
Q5. 請再次評估上述陳述的可靠性。(1) 非常不可靠 (2) 不可靠 (3) 未定 (4) 可靠 (5) 非常可靠 (96) 跳答
Q6. 阅讀此陳述後，您有多大機會在社群媒體上分享？ (1) 極不可能 (2) 不可能 (3) 未定 (4) 可能 (5) 非常可能 (96) 跳答</td><td>你听过美国前军事战略家杰克基恩说乌克兰战争是一项投资吗？ 美国只花660亿美元就能让乌克兰和俄罗斯打仗……基恩接着提到台湾也是一样，台湾公民是美国人打廉价战争的「投资」。美国冷酷精明，没有任何实际帮助台湾的意图！
Q5. 在看过事实查核结果后，请再次评估上述陈述的可靠性。(1) 非常不可靠 (2) 不可靠 (3) 未定 (4) 可靠 (5) 非常可靠 (96) 跳答
Q6. 在看过事实查核结果后，阅读此陈述后，您有多大机会在社群媒体上分享？ (1) 極不可能 (2) 不可能 (3) 未定 (4) 可能 (5) 非常可能 (96) 跳答</td></tr>
<tr><td>Post-treatment Question II</td><td>每年有打流感疫苗的要小心了！ 卫福部现在又大开后门，让高端引进国外低价疫苗，连研发都不用？ 这个无良政府为了赚钱，小老百姓可得睁大眼睛啊！ 出了事他们可不负责的事！
Q7. 請再次評估上述陳述的可靠性。(1) 非常不可靠 (2) 不可靠 (3) 未定 (4) 可靠 (5) 非常可靠 (96) 跳答
Q8. 阅讀此陳述後，您有多大機會在社群媒體上分享？ (1) 極不可能 (2) 不可能 (3) 未定 (4) 可能 (5) 非常可能 (96) 跳答</td><td>每年有打流感疫苗的要小心了！ 卫福部现在又大开后门，让高端引进国外低价疫苗，连研发都不用？ 这个无良政府为了赚钱，小老百姓可得睁大眼睛啊！ 出了事他们可不负责的事！
Q7. 在看过事实查核结果后，请再次评估上述陈述的可靠性。(1) 非常不可靠 (2) 不可靠 (3) 未定 (4) 可靠 (5) 非常可靠 (96) 跳答
Q8. 在看过事实查核结果后，阅读此陈述后，您有多大机会在社群媒体上分享？ (1) 極不可能 (2) 不可能 (3) 未定 (4) 可能 (5) 非常可能 (96) 跳答</td></tr>
</table>

Appendix C Sample Strategies

C.1 Power Analysis

We used the `pwr` R package to calcuate the adequacy of our sample size. Our study comprised a total of 1,140 agents (573 in the control group, 567 in the treatment group), with a nearly balanced allocation between conditions. Considering our model includes 22 predictors (including interaction terms and dummy-coded categorical variables), with an effect size of 0.05 and a significance level of $\alpha = 0.05$, our power analysis revealed statistical power substantially exceeding 0.90. This indicates that our experimental design has ample capacity to detect the hypothesized effects, and our sample size is sufficient to support reliable and robust statistical inferences.

Figure C.1: Power Analysis

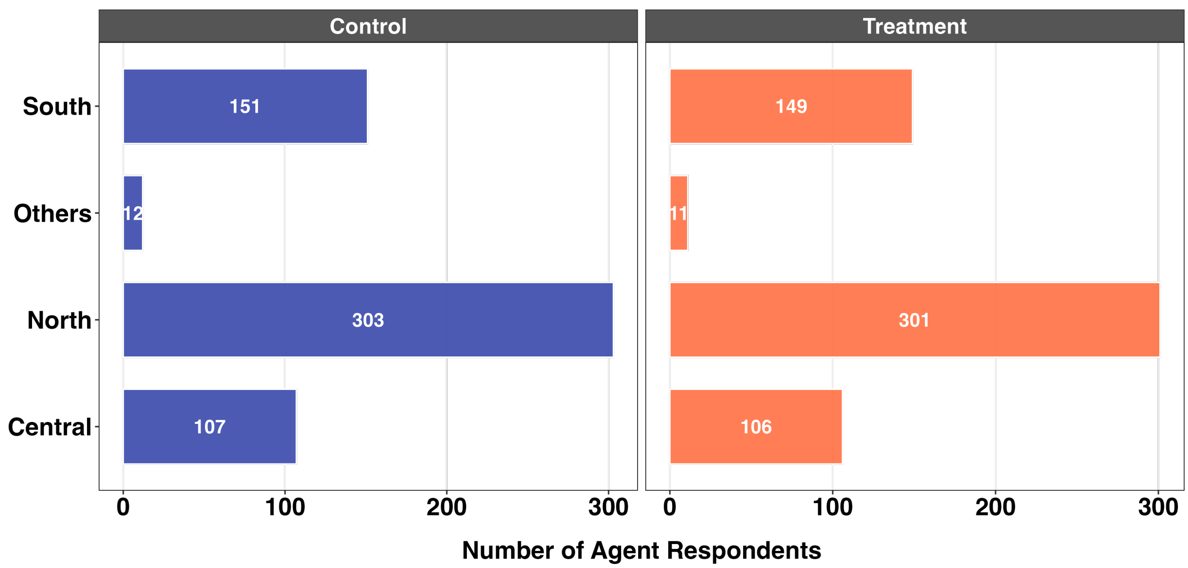


Note: Actual sample size substantially exceeds the minimum required.

C.2 Sample Distribution Across Taiwan's Regions

We recoded Taiwan's counties and cities into four main regions. Based on regional distribution proportions, we sampled a total of 1,140 respondents: 604 from the north, 213 from the central region, 300 from the south, and only 23 from other regions. The northern region includes Keelung City, Taipei City, New Taipei City, Taoyuan City, Hsinchu City, Hsinchu County, and Yilan County; the central region includes Miaoli County, Taichung City, Changhua County, and Nantou County; the southern region includes Chiayi City, Yunlin County, Chiayi County, Tainan City, Kaohsiung City, and Pingtung County; while the other regions category combines Taitung County, Hualien County, Penghu County, Kinmen County, Lienchiang County. "Others", "Skip" and "Missing" (99) are recoded as other regions.

Figure C.2: Geographic Distribution of Agent Respondents Based on the 2024 Wave of Research on China Image Survey



Appendix D Supplementary Analysis

D.1 Dimensional Analysis of pro-China Stance

We employed the blackbox function from the basicspace package to extract a pro-China stance measure based on five survey items related to China policy attitudes. To examine the model fit statistics, we assessed the proportion of variance explained by each extracted dimension. Table D.2 reveal a clearly dominant first dimension, with a single dimension explaining approximately 85.6% of the total variance in responses to the China policy scales. A second dimension contributes an additional 6.9% to the proportion of explained variance, while a third dimension adds another 4.6%. The second dimension could represent attitudes on cross-cutting issues.

Given the overwhelming dominance of the first dimension in explaining variance (85.6%), we use this dimension as our primary measure of pro-China stance. This first dimension captures the core underlying attitude structure that drives respondents' positions across the five China policy items, making it an appropriate unidimensional scale for measuring pro-China versus anti-China orientations in our subsequent analyses.

Table D.2: Blackbox Analysis: Dimension Reduction Results

	SSE	SSE Explained	Percent (%)	SE	Singular
Dimension 1	2488.56	14852.06	85.65	0.757	96.59
Dimension 2	1292.50	16048.12	6.90	0.631	34.58
Dimension 3	495.90	16844.72	4.59	0.480	28.22

D.2 Analysis of Agent Experiment Costs

Table D.3 and Figure D.3 illustrate the complete costs of using the OpenAI extension model in this experiment, which involved 1,140 agent participants (567 in the treatment group and 573 in the control group). The total experiment cost amounted to $585.35, averaging $0.51 per agent. In total, the experiment utilized 16.73 million tokens at a cost of $0.035 per 1,000 tokens.

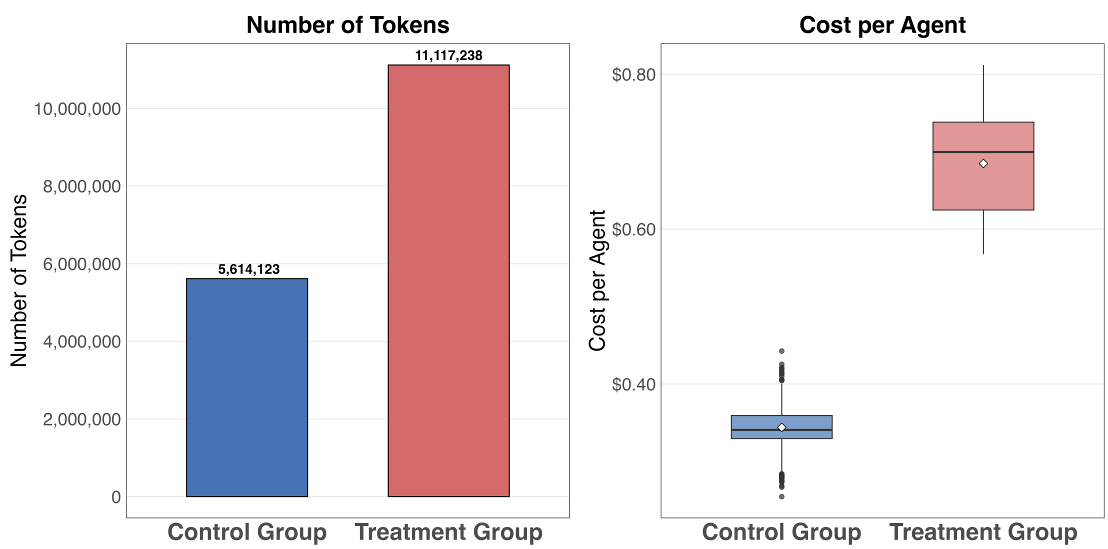
Table D.3: Cost and Token Usage Comparison by Group

Group	Agents	Total Cost ($)	Cost (%)	Total Tokens	Mean Cost ($)	Mean Tokens
Control	573	197.06	33.7%	5,614,123	0.34	9,798
Treatment	567	388.30	66.3%	11,117,238	0.68	19,607
Total	1,140	585.35	100.0%	16,731,361	0.51	14,677

Note: Cost percentage refers to the proportion of total experiment cost. Mean values represent average per agent in each group.

Treatment group costs were substantially higher than control group costs. This difference primarily resulted from the additional factchecking messages and verification processes required for the treatment group with the Survey Manager agent. These extra interactions increased token consumption, causing treatment group costs to approximately double those of the control group. This cost also reflects the varying levels of interaction complexity between the experimental conditions.

Figure D.3: Average Cost and Token Usage for Each Agent Respondent and Their Distribution



D.3 Effect Sizes of Fact-checking Interventions on Different Misinformation Scenarios

Table D.4: Effect Sizes of Fact-checking Interventions on Different Misinformation Scenarios

Misinformation Scenarios	***t*-value**	***p*-value**	**Mean Diff**	**Cohen's *d***
Belief Misinformation of US Military	-7.35	0.000***	-1.00	-1.91
Share Misinformation of US Military	-3.67	0.001**	-0.61	-0.95
Belief Misinformation Vaccine	-7.66	0.000***	-0.75	-1.99
Share Misinformation Vaccine	-3.38	0.002**	-0.36	-0.88

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Negative mean differences and Cohen's d indicates lower scores in the treatment group compared to the control group.

D.4 Full Regression of Table 3 and Table 4 in the Main Text

Table D.5: Impact of Pro-China Stance on Misinformation Susceptibility

	Misinfo. of US Army		**Misinfo. of Vaccine**	
	Model 1 (Belief)	Model 2 (Sharing)	Model 3 (Belief)	Model 4 (Sharing)
Pro-China Stance	1.47***	1.58***	1.18***	1.26***
	(0.106)	(0.187)	(0.113)	(0.174)
Party				
DPP	-0.529***	-0.440***	-0.923***	-0.853***
	(0.043)	(0.021)	(0.038)	(0.080)
New Power Party	-0.031	-0.150	-0.296	-0.341
	(0.211)	(0.301)	(0.370)	(0.172)
Green Party	-0.683*	-0.696**	-1.01*	-1.14***
	(0.270)	(0.174)	(0.373)	(0.131)
People First Party	-1.24***	-0.669***	-0.979***	-0.960***
	(0.049)	(0.052)	(0.100)	(0.122)
Taiwan Statebuilding Party	-0.586**	-0.362*	-0.998***	-0.863**
	(0.137)	(0.114)	(0.165)	(0.166)
KMT	0.510***	0.531**	0.547***	0.653**
	(0.063)	(0.156)	(0.035)	(0.135)
Taiwan People's Party	0.114*	-0.113	0.026	-0.054
	(0.047)	(0.142)	(0.132)	(0.137)
New Party	1.32*	1.45**	1.21*	1.35**
	(0.461)	(0.374)	(0.453)	(0.354)
Social Democratic Party	0.239**	-0.451***	-0.070	-0.887***
	(0.065)	(0.071)	(0.063)	(0.150)
Household Income	0.010	0.003	-0.008	-0.005
	(0.006)	(0.012)	(0.015)	(0.017)
Education				
High School	0.196	0.360	-0.105	0.130
	(0.126)	(0.189)	(0.265)	(0.508)
College and above	0.282	0.441*	0.115	0.365
	(0.137)	(0.149)	(0.160)	(0.434)
Ethnicity (Ref: Others)				
Taiwanese Minnan	-0.024	0.209	-0.129	0.136
	(0.116)	(0.188)	(0.094)	(0.181)
Taiwanese Hakka	-0.044	0.201	-0.206*	0.023
	(0.073)	(0.184)	(0.082)	(0.230)
Mainlander	0.104	0.386	-0.089	0.149
	(0.147)	(0.186)	(0.154)	(0.197)
Aboriginal	0.256	0.335	-0.482	-0.276
	(0.721)	(0.402)	(0.432)	(0.268)
Male (Ref: Female)	-0.037	0.035	0.095*	0.153*
	(0.052)	(0.094)	(0.031)	(0.059)
Political Interest	-0.075**	0.079**	-0.083**	0.082
	(0.016)	(0.024)	(0.016)	(0.036)
Region FE	✓	✓	✓	✓
Controls	✓	✓	✓	✓
Observations	981	981	981	981
R^2	0.415	0.345	0.425	0.338

Note: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Heteroskedasticity-robust standard errors in parentheses, clustered by region.

Table D.6: Difference-in-Difference Design of Factchecking Effects on Misinformation Susceptibility

	Misinfor. of US Army		Misinfo. of Vaccine	
	Model 1 (Belief)	Model 2 (Sharing)	Model 3 (Belief)	Model 4 (Sharing)
Debunking Effect	-0.136*	-0.293**	-0.080	-0.221*
	(0.054)	(0.063)	(0.049)	(0.081)
Post Treatment	0.010	0.016*	-0.023*	-0.027*
	(0.006)	(0.006)	(0.009)	(0.010)
Pro-China Stance	1.17***	1.27***	0.967***	1.02***
	(0.086)	(0.129)	(0.071)	(0.128)
Debunking Effect × Post Treatment	-1.31***	-0.805***	-1.44***	-1.04***
	(0.059)	(0.061)	(0.051)	(0.059)
Party Affiliation				
DPP	-0.385***	-0.335***	-0.694***	-0.639***
	(0.030)	(0.022)	(0.033)	(0.044)
KMT	0.336**	0.357*	0.328***	0.395*
	(0.069)	(0.137)	(0.049)	(0.129)
Taiwan People's Party	0.059	-0.097	-0.053	-0.085
	(0.031)	(0.102)	(0.083)	(0.108)
New Power Party	0.010	-0.107	-0.299	-0.388**
	(0.123)	(0.242)	(0.268)	(0.077)
Taiwan Statebuilding Party	-0.380**	-0.232**	-0.693***	-0.601**
	(0.088)	(0.058)	(0.110)	(0.109)
New Party	0.797***	0.871**	0.636***	0.771**
	(0.114)	(0.268)	(0.096)	(0.214)
Green Party	-0.652	-0.559**	-0.892	-0.874**
	(0.418)	(0.132)	(0.495)	(0.173)
People First Party	-1.27***	-0.806***	-0.946***	-1.03***
	(0.028)	(0.050)	(0.101)	(0.136)
Social Democratic Party	0.061	-0.714***	-0.186**	-1.06***
	(0.073)	(0.076)	(0.042)	(0.108)
Household Income	0.010	0.007	-0.002	0.004
	(0.009)	(0.014)	(0.016)	(0.019)
Education				
High School	-0.041	0.095	-0.220	0.032
	(0.080)	(0.147)	(0.182)	(0.321)
College and above	-0.011	0.102	-0.118	0.102
	(0.086)	(0.113)	(0.063)	(0.258)
Ethnicity (Ref: Others)				
Taiwanese Minnan	-0.025	0.172	-0.141	0.030
	(0.062)	(0.217)	(0.100)	(0.206)
Taiwanese Hakka	-0.012	0.199	-0.199	-0.023
	(0.070)	(0.230)	(0.131)	(0.251)
Mainlander	0.124	0.317	-0.029	0.110
	(0.071)	(0.206)	(0.158)	(0.209)
Aboriginal	0.093	0.110	-0.592**	-0.470**
	(0.490)	(0.191)	(0.171)	(0.103)
Male (Ref: Female)	-0.039	0.005	0.079*	0.100
	(0.073)	(0.099)	(0.026)	(0.043)
Political Interest	-0.070**	0.071***	-0.056**	0.086**
	(0.013)	(0.011)	(0.010)	(0.023)
Region FE	✓	✓	✓	✓
Observations	1,937	1,937	1,937	1,937
R^2	0.5055	0.3885	0.5086	0.39008

Note: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Heteroskedasticity-robust standard errors in parentheses, clustered by region. Panel observations from 1140 unique agents, each observed before and after treatment.

Table D.7: OLS Regressions Estimating the Effects of Pro-China Stance on Misinformation Susceptibility

	Control Group				Treatment Group			
	Misinfo. of US Army		**Misinfo. of Vaccine**		**Misinfo. of US Army**		**Misinfo. of Vaccine**	
	Model 1 (Belief)	Model 2 (Share)	Model 3 (Belief)	Model 4 (Share)	Model 5 (Belief)	Model 6 (Share)	Model 7 (Belief)	Model 8 (Share)
Pro-China Stance	1.52***	1.72***	1.22***	1.29***	1.40***	1.44**	1.14***	1.22***
	(0.16)	(0.14)	(0.12)	(0.19)	(0.18)	(0.28)	(0.14)	(0.19)
Region FE	✓	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓	✓
Observations	489	489	489	489	492	492	492	492
R^2	0.40	0.36	0.41	0.34	0.45	0.36	0.47	0.38

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Heteroskedasticity-robust standard errors in parentheses, clustered by region. Control variables include gender, ethnicity, national identity, income, party affiliation, and level of political interest. Models 1-4 represent Control Group subjects, while Models 5-8 represent Treatment Group subjects. In each group, Models 1, 2, 5, and 6 present results for Misinformation of US Army, while Models 3, 4, 7, and 8 show results for Misinformation of Vaccine.